

§13.3 Arc Length and Curvature

Homework : 13,19,21,39

(1) Arc Length

$$L = \int_a^b |r'(t)| dt$$

Why?

(i) 上學期學的 $x = f(t)$, $y = g(t)$, 參數式的距離公式.

$$\begin{aligned} L &= \int_a^b \sqrt{(dx)^2 + (dy)^2} \\ &= \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt \\ &= \int_a^b \sqrt{f'(t)^2 + g'(t)^2} dt \\ &= \int_a^b |r'(t)| dt \end{aligned}$$

Here $r(t) = \langle f(t), g(t) \rangle$.

三維的參數式同理可得類似的弧長公式.

(ii) 從物理角度看, $|r'(t)|$ 代表物體在 t 時間的瞬間速度, $|r'(t)|\Delta t$ = 小範圍

的距離 $\Rightarrow \int_a^b |r'(t)| dt$ = 代表此物體從 $t = a$ 到 $t = b$ 所走過的距離.

(2) Arc Length Function (or Distance Function)

$$s(t) = \int_a^t |r'(u)| du.$$

$$\Rightarrow \frac{ds}{dt} = |r'(t)| = \text{距離的變化率} = \text{速度}.$$

(3) 將一個 Curve 用 arc length (s) 來作參數是一個非常有用的想法和技巧.

(如此的表達方式不隨著不同的座標系統而改變.)

(4) Unit Tangent vector : $T(t) = \frac{r'(t)}{|r'(t)|}$.

(5) Curvature(曲率) : a measure of how quickly the curve changes direction at a given point.

Definition : $\kappa = \text{曲率} = \left| \frac{dT}{ds} \right|$

$$\text{定理 (i)} \quad \kappa = \left| \frac{dT}{ds} \right| = \left| \frac{dT/dt}{ds/dt} \right| = \left| \frac{T'(t)}{r'(t)} \right| = \frac{|r'(t) \times r''(t)|}{|r'(t)|^3}$$

(ii) Given a plane curve $y = f(x)$, then its curvature κ at a given point

x is

$$\kappa(x) = \frac{|f''(x)|}{\left[1 + (f'(x))^2\right]^{3/2}}$$

Proof of (i) :

$$r' = |r'|T = \frac{ds}{dt}T \quad (1)$$

$$\Rightarrow r'' = \frac{d^2s}{dt^2}T + \frac{ds}{dt}T' \quad (2)$$

(1)×(2)

$$\Rightarrow r' \times r'' = \left(\frac{ds}{dt}\right)^2 (T \times T').$$

$$\Rightarrow |r' \times r''| = \left(\frac{ds}{dt}\right)^2 |T| |T'| = \left(\frac{ds}{dt}\right)^2 |T'|$$

$$\Rightarrow |T'| = \frac{|r' \times r''|}{\left(\frac{ds}{dt}\right)^2} = \frac{|r' \times r''|}{|r'|^2}$$

$$\Rightarrow \kappa = \frac{|T'|}{|r'|} = \frac{|r' \times r''|}{|r'|^3}$$

(6) Principal unit normal vector $N(t)$.

$$N(t) = \frac{T'(t)}{|T'(t)|}$$

(7) Binormal Vector $B(t)$.

$$B(t) = T(t) \times N(t)$$

